

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

expr.l

```
%{
extern int yyval;

%}

%%

[0-9]+ { yyval=atoi(yytext); return NUMBER; }

\n { return NL ;}

. { return yytext[0]; }

%%

int yywrap()
{
return 1;
}
```

expr.y

```
%{
#include<stdio.h>
#include<stdlib.h>

%}

%token NUMBER NL

%left '+' '-'
%left '*' '/'

%%

stmt : exp NL { printf("Value = %d\n",$1); exit(0); } ;
exp : exp '+' exp { $$=$1+$3; }
    | exp '-' exp { $$=$1-$3; }
    | exp '*' exp { $$=$1*$3; }
    | exp '/' exp { if($3==0) { printf("Cannot divide by 0"); exit(0); } else $$=$1/$3; }
    | '(' exp ')' { $$=$2; }
    | NUMBER { $$=$1; } ;
```

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```
%%  
  
int yyerror(char *msg)  
  
{  
    printf("Invalid Expression\n");  
    exit(0);  
}  
  
#include "lex.yy.c"  
  
main ()  
{  
    printf("Enter the expression\n");  
    yyparse();  
}
```

OUTPUT

```
student@compalc17-OptiPlex-3010:~/Desktop/22C007$ flex expr.l  
student@compalc17-OptiPlex-3010:~/Desktop/22C007$ bison -d expr.y  
student@compalc17-OptiPlex-3010:~/Desktop/22C007$ gcc expr.tab.c -o calculator -lfl  
expr.tab.c: In function 'yyparse':  
expr.tab.c:978:16: warning: implicit declaration of function 'yylex' [-Wimplicit-function-declaration]  
  978 |         yychar = yylex ();  
      |                   ^~~~~~  
expr.tab.c:1155:7: warning: implicit declaration of function 'yyerror'; did you mean 'yyerrok'? [-Wimplicit-function-declaration]  
 1155 |         yyerror (YY_("syntax error"));  
      |         ^~~~~~  
      |         yyerrok  
expr.y: At top level:  
expr.y:23:1: warning: return type defaults to 'int' [-Wimplicit-int]  
   23 | main ()  
      |      ^  
student@compalc17-OptiPlex-3010:~/Desktop/22C007$ ./calculator  
Enter the expression  
3*7+1  
Value = 22  
student@compalc17-OptiPlex-3010:~/Desktop/22C007$
```

CONCLUSION: YACC program to parse an expression for a given grammar was implemented and executed successfully.